Concept of variation

1

a When x increases, y increases. When x decreases, y decreases.

b When x increases, y decreases. When x decreases, y increases.

2

a Direct

b Inverse

3 It is doubled

4

a Direct

b Inverse

c Inverse

Direct variation

5

a $y = 30$

b $y = 15$

c 3

6 $q = \frac{2}{p^2}$

7 $t = \frac{48}{s}$

8 The distance from a storm varies directly as the number of seconds between seeing lightning and hearing thunder.

9 The speed varies directly as the distance travelled, and inversely as the time taken.

10 The centripetal force of an object varies directly as its mass and the square of its velocity, and inversely as the radius of the circle it moves along.

11

a $k = 0.29$

b $M = 0.29x^3$

c $M = 148.48$ g

12

a $k = 1.73$

b $A = 1.73s^2$

c $A = 15.57 \text{ cm}^2$

13

14

a $k = \dfrac{2}{289}$

b $n = 11$

Inverse variation

15 When one quantity decreases the other quantity increases, and vice versa.

16 $r = \dfrac{k}{a}$

17

a $y = 9$

b 18

18

a $k = 0.000368$

b $y = \dfrac{0.000368}{x^2}$

19

a Yes

b No

c Yes

d Yes

e No

f No

g No

h Yes

i No

j Yes

k Yes

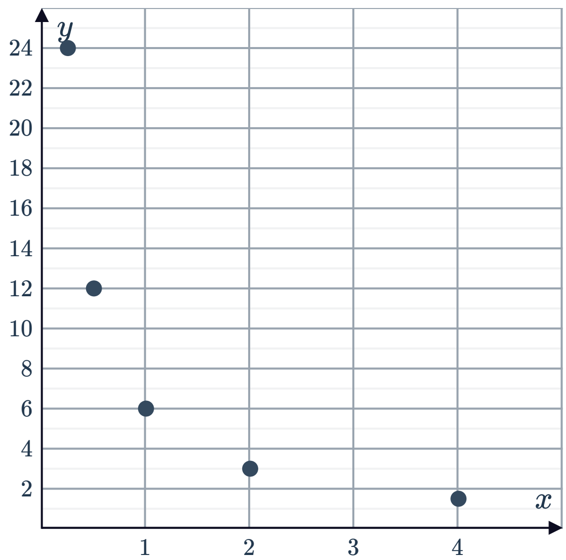
l No

20

a

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y	24	12	6	3	$\frac{3}{2}$

b



21

a $k = 200$

b $a = \dfrac{200}{x}$

c $a = 40$

22 **B**

23 $r = \frac{5}{n}$

24

a 375

b $s = \frac{125}{2}$

c $s = \frac{125}{4}$

25

a $k = 1220$

b $t = 40$